

## RAGs to Riches: A Reimplementation of RAG for Knowledge-Intensive NLP Tasks

### Introduction

The main problem we are trying to solve is open-domain question answering, where models have to answer factual questions using broad world knowledge. Closed-book models, like T5, rely solely on knowledge stored in their parameters (parametric memory), which is inadequate for answering questions that require external or updated information. Hence, closed-book models are likely to hallucinate. Our project re-implements a simplified version of *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks* by Lewis et al. The RAG model introduced in the paper combines a retriever with a sequence-to-sequence generator. Instead of generating answers from parametric knowledge only, RAG models retrieve the relevant passages from an external corpus (non-parametric memory) and, based on them, generate an answer. Thus, RAG is a combination of parametric and non-parametric memory.

### Chosen Result

In our project, we aimed to reproduce the results of Table 1 for the Natural Questions dataset. The table compares the closed-book T5-base model and the open-book models: DPR (Dense Passage Retriever), RAG-Token, and RAG-Sequence. We chose this table in particular because it validates the paper's central claim: retrieval-augmented generation effectively grounds open-domain question answers in factual evidence, overcoming closed-book model limitations. Since our computations were limited by Google Colab resources, our main goal was not to get the exact same results, but to reproduce the general trend, which is that RAG variants consistently outperform both DPR and T5 baselines.

| Model       |                | NQ          |
|-------------|----------------|-------------|
| Closed Book | T5-11B [52]    | 34.5        |
|             | T5-11B+SSM[52] | 36.6        |
| Open Book   | REALM [20]     | 40.4        |
|             | DPR [26]       | 41.5        |
| RAG-Token   |                | 44.1        |
| RAG-Seq.    |                | <b>44.5</b> |

**Figure 1.** Open-Domain QA Test Scores

### Methodology

To evaluate the impact of non-parametric memory, we developed a pipeline that balances the architectural complexity of RAG with the computational limits of our environment. Our re-implementation approach is defined by the following:

- **Model Comparison:** We analyzed four configurations: T5-base as a closed-book baseline, DPR as a retrieval-only baseline, and two manually implemented variants: RAG-Token and RAG-Sequence.
- **Dataset:** We used Natural Questions (NQ), a dataset of real-world Google user questions paired with short gold answers.
- **Retrieval and Indexing:** We used DPR to encode questions and passages into dense vectors stored in a FAISS index to retrieve the top-k passages. The original paper uses a Wikipedia index

as the retrieval source. However, due to Colab storage limitations, we utilized SQuAD context passages as the closest resource-efficient corpus to Wikipedia. While SQuAD is less broad than Wikipedia, it served as a representative knowledge base for our model scale.

- **Generation:** We used BART-base instead of BART-large to optimize for Colab memory constraints, pairing each retrieved passage with a question before passing it into the model. We implemented two distinct generation strategies:
  - **RAG-Sequence:** each of the top-k passages is used to generate a full answer; the model then picks the one that is most likely to be correct.
  - **RAG-Token:** evaluates all top-k passages for each token produced, allowing the model to switch between multiple contexts mid-sentence to synthesize a factual response.
- **Evaluation:** While the original paper uses Exact Match, we used token-level F1. Exact Match is too strict, so it would have been impossible for us to get any non-zero value. Hence, we used F1 for partial credit on overlaps between prediction and golden answer, even if not exactly the same.

## Results & Analysis

Although our results did not exactly match the original scores due to computational constraints, our reproduction successfully confirmed the reported trends. T5-base performed the worst at 8.1 F1, as its reliance on parametric memory alone is insufficient to cover the broad knowledge required for Natural Questions. While DPR outperformed T5 at 13.73 F1, it lacked the generative capabilities of the RAG variants. Both RAG models significantly outperformed the baselines, with RAG-Token achieving the highest accuracy at 33.8 F1. This is because RAG-Token is more flexible even when a retriever pulls suboptimal passages since each token can still be produced using other context from the top-k passages. RAG-Sequence (31.44 F1) proved more sensitive to retrieval quality because it generates the entire answer from a single passage.

| Book        | Model                             | Paper NQ (EM) | Our NQ (F1) |
|-------------|-----------------------------------|---------------|-------------|
| Closed Book | T5 (paper: T5-11B, ours: T5-base) | 34.5          | 8.1         |
| Open Book   | DPR                               | 41.5          | 13.73       |
| Open Book   | RAG-Token                         | 44.1          | 33.8        |
| Open Book   | RAG-Sequence                      | 44.5          | 31.44       |

**Figure 2.** Paper results (Exact Match) vs. our re-implementation (Token-level F1) on NQ dataset

We also conducted a small experiment to analyze the influence of the retrieval depth,  $k$ , on the quality of the results. Increasing  $k$  improved the results of RAG-Token, but after  $k=5$ , the improvement was not significant. This suggests that the first few passages provided sufficient data, while additional passages were either redundant or noisy. In contrast, RAG-Sequence results remained constant at approximately 31.48 F1 because the model always prioritized the single highest-probability passage to generate the entire response. These findings highlight that while deeper retrieval benefits token-level synthesis, sequence-level generation is limited by the quality of its primary source.

| k  | RAG-Token F1 | RAG-Sequence F1 |
|----|--------------|-----------------|
| 1  | 32.1         | 31.48           |
| 3  | 33.92        | 31.48           |
| 5  | 33.8         | 31.48           |
| 10 | 33.89        | 31.48           |

**Figure 3.** Performance impact of retrieval depth (k) on RAG variants

### **Reflections**

RAG is conceptually simple, but difficult to reproduce at scale. The results heavily depended on the quality of the retrieval corpus. The main takeaway is that retrieval quality is the main bottleneck for open-domain QA: good retrieval provided sentences with significant amounts of evidence, so the produced answers were strong, but when the retrieved passages were weak, even a powerful generator like BART could not fully recover.

RAG-Token ended up being more robust in our setup than RAG-Sequence. RAG-Token provided some flexibility because it combined evidence across all documents at token-level, while RAG-Sequence was more dependent on document retrieval quality. Our retrieval depth experiment supported it, as increasing k improved the quality of generated answers up to the point when the gains became insignificant as extra passages were repetitive or noisy.

Future works would include using the entire Wikipedia corpus, scaling the retriever to Wiki-DPR, using BART-large instead of BART-base, and fine-tuning the retriever together with the generator. Potentially, we could explore better grounding methods to force the generator to rely more on retrieved evidence as well.

### **References**

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